Engineering Mathematics - II

Case Study

Stokes' Theorem

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Introduction

Stokes' theorem, also known as the *Kelvin-Stokes theorem* after Lord Kelvin and Sir George Gabriel Stokes, the *fundamental theorem for curls* or simply the *curl theorem*, is a theorem in vector calculus.

Given a vector field, the theorem relates the integral of the curl of the vector field over some surface, to the line integral of the vector field around the boundary of the surface.

The exact statement goes as follows:

Let S be an open surface bounded by a closed curve C, and $F=f_1I+f_2J+f_3K$ *any continuously differentiable vector point function, then*

$$\iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

where n is a unit external normal at any point of S.

This classical Kelvin-Stokes theorem is a special case of the actual, generalised Stokes' theorem, which is a statement about the integration of differential forms, generalising several theorems from vector calculus. The special case as defined above is the 1+1 dimensional case, for a 1-form. It is a statement about vector fields and is a good introductory pathway towards the general form.

An interesting thing to note is that Green's theorem is actually a special case of the Kelvin-Stokes theorem (which from here on will simply be referred to as *Stokes' theorem* or *Curl theorem*). Or, we can say that Stokes' theorem is a generalisation of Green's Theorem.

So what is Green's Theorem?

Let C be a positively oriented, piecewise-smooth simple closed curve in the plane, and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

If we simplify our function $F=f_1\mathbf{I}+f_2\mathbf{J}+f_3\mathbf{K}$ *to* $F=P\mathbf{I}+Q\mathbf{J}$, that is, if we reduce the dimensions from three to two, then Stokes' theorem will take the form of Green's Theorem. Hence, it is shown how Green's theorem is a specific case of Stokes' theorem.

It is important to note that all these theorems (including the divergence theorem as well) are all special cases of the generalised Stokes' theorem.

History

The origins of Stokes' theorem, Green's theorem and Gauss' Divergence theorem can all be traced back to the late 17th century. However, the theorems as we know them today did not appear explicitly until the 19th century.

In 1847, Sir George Gabriel Stokes published his paper *On the Theory of Oscillatory Waves.* In this paper, he introduced the concept of the circulation of a fluid around a closed curve, and established a connection between the circulation and the curl of the velocity field. This idea is what would go on to develop and become Stokes' theorem.

The first time Stokes theorem appeared in print was in the year 1854. George Stokes had for several years been setting the Smith's Prize Exam at Cambridge, and in the February 1854 examination, question #8 was the following:

If X, Y, Z be functions of the rectangular coordinates x, y, z, dS an element of any limited surface, l, m, n the cosines of the inclinations of the normal at dS to the axes, ds an element of the boundary line, shew that

$$\iint \left\{ l\left(\frac{\partial Z}{\partial y} - \frac{\partial Y}{\partial z}\right) + m\left(\frac{\partial X}{\partial z} - \frac{\partial Z}{\partial x}\right) + n\left(\frac{\partial Y}{\partial x} - \frac{\partial X}{\partial y}\right) \right\} dS = \int \left(X\frac{dx}{ds} + Y\frac{dy}{ds} + Z\frac{dz}{ds} \right) ds$$

... the single integral being taken all around the perimeter of the surface.

It does not seem to be known if any of the students proved the theorem. However, the theorem had already appeared in a letter of William Thomson (Lord Kelvin) to Stokes on July 2, 1850. The left hand expression of the theorem had appeared in two earlier works of Stokes.

However, the first published proof of the theorem seems to have been in a monograph of Hermann Hankel in 1861. Hankel gives no credit for the theorem, only a reference to Riemann with regard to Green's theorem, which is a theorem he calls well-known and makes use of in his own proof of Stokes' result.

Hankel's proof required the surface to be given explicitly as z=z(x,y). A somewhat different proof, without that requirement, was sketched in Thomson and Tait's *Treatise on Natural Philosophy* (1867) without reference.

In 1871, Clerk Maxwell wrote to Stokes, asking about the history of the theorem. Evidently, Stokes answered him, since in Maxwell's 1873 *Treatise on Electricity and Magnetism,* there appears the theorem with the reference to the Smith's Prize Exam. In fact, two of the four Maxwell equations, namely the Maxwell-Faraday equation and the Ampère-Maxwell law, involve curls of three-dimensional vector fields, and their differential and integral forms are actually related by Stokes' theorem.

By the 1890's, Stokes' theorem was appearing consistently in the analysis texts of many different authors. This theorem was always attributed to Stokes himself, which is not something that one could say for some of the other theorems and the mathematicians or physicists they were named after.

The theorem was further worked on and developed, and with the understanding and application of it, there were crucial advancements in several fields. Stokes' theorem was ultimately formulated in its modern form by Élie Cartan in 1945, following earlier work on the generalisation of the theorems of vector calculus by Vito Volterra, Édouard Goursat, and Henri Poincaré.

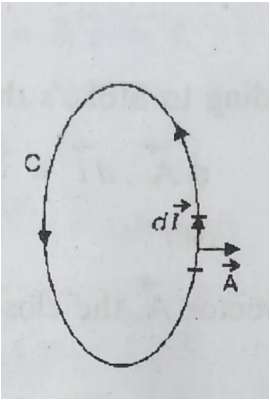
There hasn't been any significant updates or revisions to Stokes' theorem after that. For the last 80 years, it has remained a well-established and fundamental result that has many real-world applications.

Proof

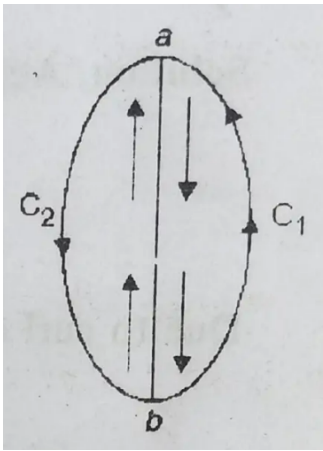
Let A be the vector point function acting on a surface S, enclosed by a closed curve C. The line integral of A vector along a closed curve is given by

$$\oint_C \vec{A}.\overrightarrow{dl}$$

Where dl is the length of a small element of the path, as shown in the below figure.



Now, let's divide the area enclosed by the closed curve C into two equal parts by drawing the line ab. We now have two closed curves C1 and C2.



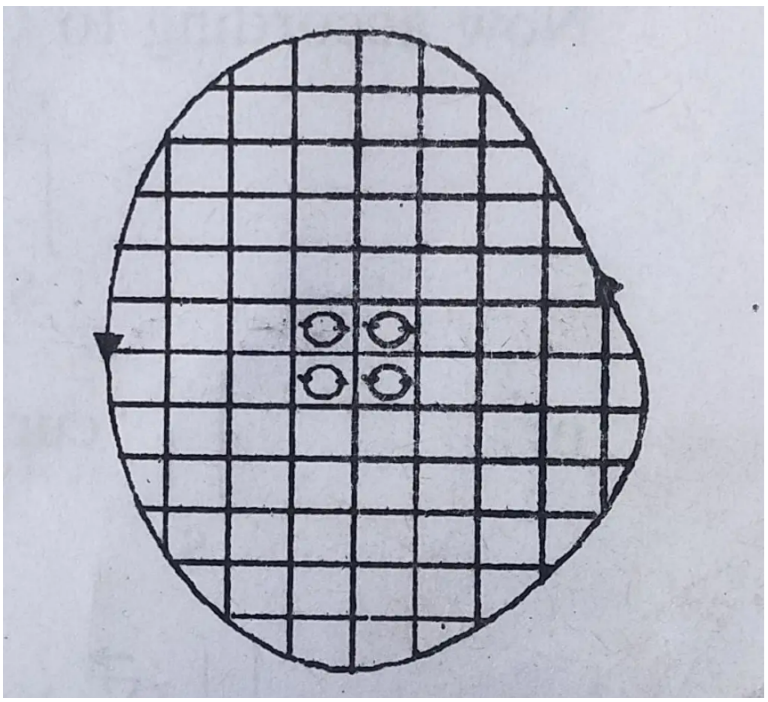
Therefore, the line integral of vector A along a closed curve C can be written as

$$\oint_{C1} \vec{A}.d\vec{l} = \oint_{C2} \vec{A}.d\vec{l} + \oint \vec{A}.d\vec{l} \quad \ldots \quad \ldots \quad \ldots \ldots \textbf{(1)}$$

If the area enclosed by the curve C is divided into a large number of small areas such as dS_1, dS_2, dS_3, ..., dS_n bounded by the curves C_1, C_2, ..., C_n.

$$\oint_{C} \vec{A}.d\vec{l} = \sum \oint_{Cn} \vec{A}.d\vec{l} \ldots \ldots \ldots \ldots (2)$$

The corresponding diagram is given below.



According to the definition of curl,

$$\oint_{Cn} \vec{A}.d\vec{l} = Curl\ \vec{A}.d\ \overrightarrow{S_n} \ldots\ldots\ldots\ldots (3)$$

Put this value in (2), and we get,

$$\oint_{C} \vec{A}.d\vec{l} = \sum \oint_{Cn} \vec{A}.d\vec{l} = \sum Curl\ \vec{A}.d\overrightarrow{S_n} \ldots\ldots\ldots\ldots (4)$$

where dS_n is the surface area in the case under consideration. When we take the limit of the above summation, it becomes an integral.

$$dS \to 0, \sum Curl\ \vec{A}\,.d\overrightarrow{S_n} = \iint_{S} Curl\ \vec{A}.d\vec{S}$$

Hence, equation 4 can be written as

$$\oint \vec{A}.d\vec{l} = \iint Curl\ \vec{A}.d\vec{S}\ \ or\ \oint \vec{A}.d\vec{l} = \iint (\vec{\nabla} \times \vec{A})\,.d\vec{S} \ldots\ldots\ldots (5)$$

which is Stokes' theorem.

Thus Stokes' theorem has been successfully proved.

General Applications

Here are some of the general applications of Stokes' theorem, a few of the areas where it has been and is still being used.

1) Helmholtz's theorem is a theorem in fluid dynamics that is derived from Stokes' theorem. It characterises vortex-free vector fields. This theorem provides an explanation as to why the work done by a conservative force in changing an object's position is path independent, which is an important result in physics.

2) The accurate prediction of wind turbine aerodynamics is still challenging. Computational Fluid Dynamics (CFD) codes suffer from high computational cost and rapid artificial dissipation of vortical structures in the wake of wind turbines and rotorcrafts. Prescribed and free wake models are computationally more efficient, yet fail to predict accurately separated and stalled flow along the blade, as available two-dimensional airfoil polar data are used directly or modified semi-empirically to account for three-dimensional effects.

Some hybrid methods have been developed for wind turbines and rotorcrafts, combining a near-field Navier-Stokes (NS) analysis with a far-field representation by potential and vortex methods, thus reducing both artificial dissipation and computational cost compared to a full domain NS analysis. The linking quantity within hybrid methods is the spanwise bound circulation Γ. In general, Γ inside the NS zone is determined by the Kutta-Zhukovsky (K-Z) lift theorem from sectional blade loads. However, the K-Z lift theorem is strictly valid only for 2D inviscid and incompressible flow. A recently developed parallelized coupled Navier-Stokes/Vortex-Panel solver (PCS) determines Γ by integrating along closed contour lines around each section of blade according to Stokes' theorem and without further assumptions. This work presents the derivation of a viscous lift theorem, the validation in 2D, and the application to a parked wind turbine blade as well as a comparison with measured wind tunnel data and a vortex line method (VLM).

3) Stokes' theorem can be compared to holography on some levels, but in its most basic form, it works with fluids or fluid-like substances. It is used to figure out how

viscous a fluid like oil is, by letting a sphere of known diameter fall freely in the liquid. Following the initial acceleration, the external drag on the surface and the buoyancy both act upwards and in opposite directions to the motions. The theorem also aids in the detection of silt settling in freshwater.

4) In magnetic field analysis, Stokes' theorem is employed to calculate magnetic flux and relate it to the circulation of the magnetic field. This is used in the design and analysis of magnetic systems, such as electric motors, transformers, and magnetic sensors.

Applications in the Field of IT

Stokes' theorem doesn't really have many direct applications in the field of Information Technology as a standalone theorem, but its principles and concepts alongside other conclusions and theorems do have connections to IT.

1) The justification for the equivalence of the differential and integral forms of the Maxwell-Faraday Equation and the Ampère-Maxwell law was provided by Stokes' theorem. These equations form the foundation of electromagnetism and electromagnetic waves, and as we know, without electromagnetic waves such as radio waves and light waves, communication itself is impossible. Broadcasting of radio signals and TV station signals, night vision cameras that use infrared rays, internal photography using x-rays for detection of injuries are all the various applications of electromagnetic waves; without the discovery and proper understanding of these waves through Maxwell's laws, a lot of the very important things in the world would simply not exist. Stokes theorem played an integral role in forming these equations in the first place.

2) Mathematical models and simulations are often used by IT professionals in order to solve complex problems. Mathematics as a whole is crucial to the IT field when it comes to real-life, practical problems and solutions. Stokes' theorem, along with other important mathematical concepts, provides a foundation for constructing these models and simulating real-world phenomena accurately. For example, simulating the transfer of heat in electronic devices can involve applying principles related to Stokes' theorem, and this helps in providing more energy efficient devices which sustain their lives for a longer period.

3) In the area of Statistics and Data Science, visualisation of data is crucial. It is important to represent complex data in a visually pleasing and simple form for proper analyses. In some cases, the visualisation of vector fields may require application of principles related to Stokes' theorem.

These are but a few of the areas where Stokes' theorem, alongside other results and concepts, plays an important role.

Sample Problem

Let us now see how Stokes' theorem is actually used in a mathematical problem. Often, real-life problems are put forward in a mathematical perspective to use these theorems easily and find solutions for them.

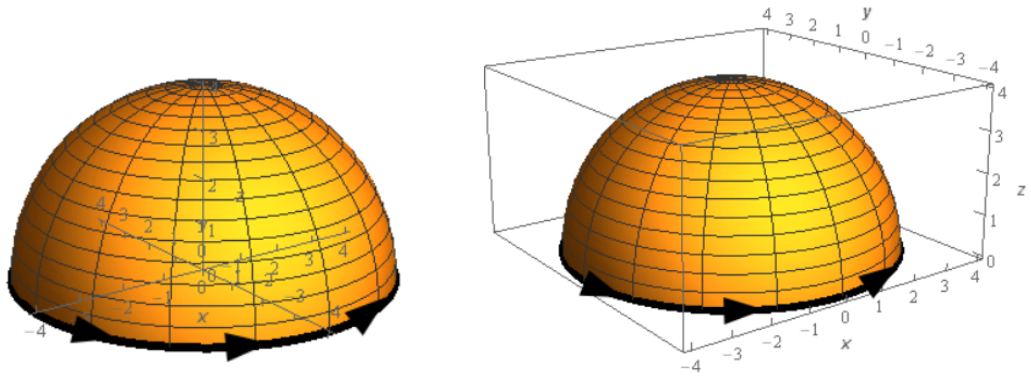
Use Stokes' theorem to evaluate

$$\iint_S \text{curl}\vec{F} \cdot d\vec{S} \text{ where } \vec{F} = y\,\vec{i} - x\,\vec{j} + yx^3\,\vec{k}$$

*where **S** is the portion of the sphere of radius 4 with z >= 0, and the upwards orientation.*

Solution

Let's start off with quick sketches of the surface we are working with in this problem. One sketch has traditional axes, and the other contains *box* axes to help visualise the surface.

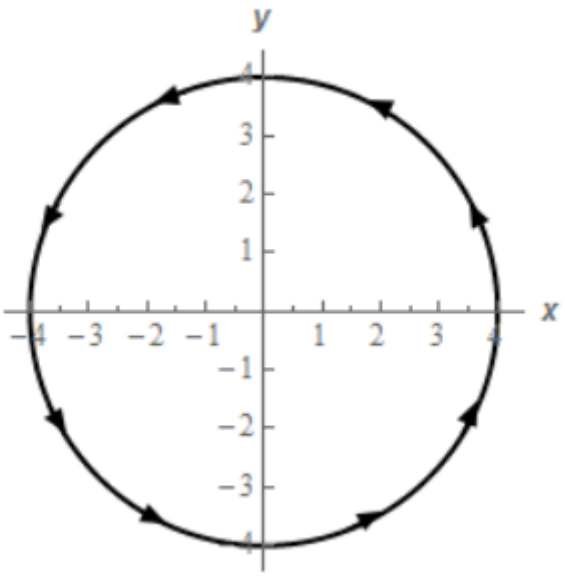


The orientation of the surface is upwards, so all the normal vectors will be pointing upwards. So, if we walk along the edge of the surface, that is, the curve *C*, in the direction indicated, with our head pointed away from the surface (that is, in the same direction as the normal vectors), our left hand will be over the region. Therefore, the direction indicated in the sketch is the positive orientation of *C*.

Let us thus use Stokes' theorem.

$$\iint_S \text{curl}\vec{F} \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{r}$$

Since *C* is just the curve along the bottom of the upper half of the sphere, we can see that *C* will in fact be the intersection of the sphere and the *xy*-plane (which is *z*=0). So, *C* is just the circle of radius 4. Looking at the sphere from above, we will get the sketch of *C.*



The parameterisation of *C* is given by,

$$\vec{r}\,(t) = \langle 4\cos t, 4\sin t, 0\rangle \qquad 0 \le t \le 2\pi$$

The *z* component of the parameterisation is 0, because *C* lies in the *xy*-plane.

Since we know we'll eventually need to do the line integral, let us calculate the following dot product.

$$\begin{aligned}\vec{F}\left(\vec{r}\left(t\right)\right) \cdot \vec{r}'\left(t\right) &= \left\langle 4\sin t, -4\cos t, 256\sin t\cos^3 t\right\rangle \cdot \left\langle -4\sin t, 4\cos t, 0\right\rangle \\ &= -16\sin^2 t - 16\cos^2 t \\ &= -16\end{aligned}$$

The parameterisation of *C* has to be plugged into the vector field, not *C* itself.

Finally, we can evaluate the integral using Stokes' theorem.

$$\begin{aligned}\iint_S \mathrm{curl}\vec{F} \cdot d\vec{S} &= \int_C \vec{F} \cdot d\vec{r} \\ &= \int_0^{2\pi} -16\, dt = \boxed{-32\pi}\end{aligned}$$

This is the final answer.

Conclusion

Stokes' theorem is a fundamental tool in mathematical physics, and its influence and applications go even beyond that, in areas which seem insignificant and trivial but in reality make up important aspects of our lives. It has many practical applications on its own and also alongside other results. Ultimately, it is a fundamental and powerful tool which has expanded our understanding of the world in a mathematical and realistic perspective.

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